

Team Contribution Report

Characterizing Electricity Demand and Marginal CO₂ Emissions During Hazardous Grid Events

Data Mining Group Project - Spring 2026

Group - 4:

Atharva Zodpe - Website Development + Content Structuring

- Built and deployed the full project website on GitHub Pages, implementing all required milestone sections: Introduction, Proposal, Data Exploration, Research Questions, Models, and Team.
- Designed the site's navigation, layout, and section hierarchy to ensure logical flow from problem framing through methodology to results.
- Structured the Models Implemented section with consistent per-model formatting covering rationale, assumptions, hyperparameters, challenges, and metrics.
- Integrated all team-generated visualizations, tables, and written analysis into the website in a coherent, accessible format.
- Maintained the repository and coordinated milestone submission artifacts including GitHub URL and required deliverables checklist.
- Integrated all four model result visualizations into the website's Models Implemented section, ensuring each model's metrics, assumptions, and challenge write-ups were consistently formatted and rubric-compliant across both the site and the PDF report.

Moulik Kumar - Dataset Acquisition + Integration

- Identified and evaluated the WattTime API and EIA-930 API as primary data sources, justifying API-first collection over static datasets for reproducibility and recency.
- Designed the integration approach to merge emissions and demand time-series into a unified schema, including timezone normalization to UTC and region-to-balancing-authority mapping.
- Defined the preprocessing specification: forward-fill for gaps under 30 minutes, linear interpolation for 30 min–2 hr gaps, and exclusion of events with over 10% missing data.

- Specified outlier handling — IQR method with domain thresholds (demand > 0, emissions < 1500 lbs/MWh), 99th-percentile capping, and manual review of the top 20 anomalies.
- Contributed to the hazardous event catalog design, coordinating the source list and structured metadata schema (event type, severity, timestamps, affected customers).
- Designed the event-level feature dataset (`event_features.csv`), defining the aggregation schema and z-score anomaly detection logic (demand $z > 2$ AND MOER $z > 1$, rolling 24-hour window) used to identify and merge contiguous event windows into discrete model-ready records.

Pratik Patil - Exploratory Analysis + Feature Engineering

- Conducted primary exploratory data analysis on the 2.1M-record cleaned dataset, inspecting distributions, temporal patterns, and regional variation across CAISO, PSCO, and ERCOT.
- Produced all ten EDA visualizations: time-series, distribution analysis, correlation heatmap, seasonal decomposition, scatter plots, duration histograms, Q-Q plots, hourly heatmaps, cumulative emissions, and pair plots.
- Proposed and implemented the feature engineering logic: 24-hour rolling mean and standard deviation of MOER, lag features at 1-hour/6-hour/24-hour intervals, and deviation-from-baseline metrics.
- Wrote the analytical insight for each visualization, identifying findings such as the pre-restoration emissions spike, weak demand-emissions coupling during events ($r=0.38$), and duration as the dominant driver of total emissions impact ($r=0.84$).
- Contributed to research question framing, grounding each of the ten questions in patterns observed during exploratory analysis.
- Implemented all four models end-to-end (`models.py`, `LogisticRegressionModel.ipynb`) — K-Means ($k=4$, silhouette=0.584), Logistic Regression (CV AUC=0.65, StratifiedKFold), Linear Regression ($R^2=0.237$, RMSE=105.4, `is_event` indicator added post-residual inspection), and FP-Growth (34 rules, max lift=1.77, min_support=0.30) — including hyperparameter tuning, evaluation metrics, and result interpretation for each.